

- RÉMI DI GUARDIA, THOMAS EHRHARD, AND CLAUDIA FAGGIAN, *Quantum Bayesian networks: compositionality and typing via linear logic*.
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Summary. Quantum Bayesian networks [6] provide a mathematical formalism to describe causal relations, to analyse correlations, and to predict the probabilities of measurement outcomes, in systems involving both *classical and quantum* data. They generalise Pearl’s Bayesian networks [9]—prominent graphical models for classical probabilistic reasoning and inference.

This abstract aims to bring compositional principles and a typing discipline into this setting; see the conference version [3] for details of definitions and results. A key feature of our compositional semantics is that when all causes are classical, it coincides with the standard factor-based semantics of Bayesian networks, while in the purely quantum case it reduces to tensor networks. We then propose a typed formalism based on proof-nets from linear logic, where types ensure well-behaved composition of systems, and which we prove sound and complete with respect to quantum Bayesian networks.

Background. Pearl’s *Bayesian networks* [8, 9] provide a framework for reasoning under conditions of uncertainty and partial knowledge, with a wide range of applications from statistics to epidemiology, economics and computer science. Bayesian networks have a *dual nature*, serving both as probabilistic models for inference, and as causal models specifying the connections between observed data. When reasoning on *quantum systems*, this framework is not general enough to account for entanglement and non-local correlations. The development of quantum causal models—see *e.g.* [1] and references therein—is an active research area across quantum information and the foundations of quantum theory. Here, we focus on *quantum Bayesian networks*, a generalisation of Bayesian networks introduced by Henson, Lal and Pusey [6]. They provide a framework to describe causal relations and to predict the probabilities of measurement outcomes in systems with both classical and quantum data. The formalism builds on work by Leifer and Spekkens [7], with as perspective *quantum theory as a theory of inference*. Quantum theory is indeed probabilistic, as it is concerned with predicting the probabilities of measurement outcomes on a system. Hence, it can be framed as a *probabilistic inference over models with both classical and quantum data*.

EXAMPLE 1. The directed acyclic graph (DAG) in Figure 1 describes the well-known Bell experiment. Quentin prepares a pair of (possibly entangled) qubits, sending one to Alice and the other to Bob. Alice randomly performs one of two possible measurements, by flipping a coin X . Bob also performs a measurement, by flipping a coin Y . The result of the experiment is $\Pr(a, b \mid x, y) = \Pr(a, b, x, y) / \Pr(x, y)$ *i.e.* the probability that the (classical) outcomes of Alice’s and Bob’s measurements are a and b , given outcomes x for X and y for Y .

Bayesian networks. In the theory of Bayesian networks, the causal structure is encoded by a *DAG* where nodes represent random variables and edges conditional dependencies. The semantics of a Bayesian network is a probability distribution. Bayesian networks are a compact description of large probability distributions, with efficient inference algorithms to answer queries on these distributions without fully constructing them. These rely on *factors*, a generalisation of conditional probability distributions, whose operations *inherently share variables*, allowing for tractability and efficiency.

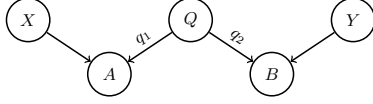


FIGURE 1. Bell set-up (from [6])

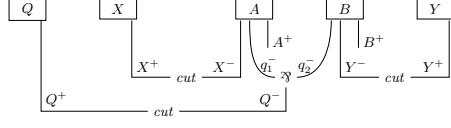


FIGURE 2. Bell set-up as a quantum proof-net

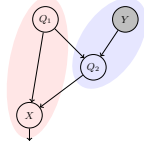


FIGURE 3. Compositionality

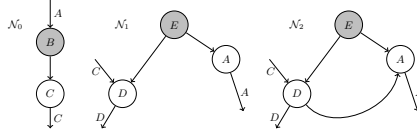


FIGURE 4. Modularity

Quantum Bayesian networks. Quantum Bayesian networks are still an emerging field, not as developed as their classical counterparts. A crucial missing feature is *compositionality*, the ability to compute the semantics through intermediate, partial computations—*i.e.* computing a model’s denotation as a function of its subparts. Consider the DAG in Figure 3: nodes X and Y produce a classical output, while nodes Q_1 and Q_2 have a quantum nature. A natural question is: *Can we compute the semantics of the model in terms of sub-components—e.g. the highlighted sub-graphs?* The approach in [6] does not adapt well (see [6, page 12]).

A closely related question concerns *modularity*: *When can causal descriptions of systems as subparts be used to construct larger models?* Consider the DAGs in Figure 4: $\mathcal{N}_0$ awaits an input A and outputs C , $\mathcal{N}_1$ and $\mathcal{N}_2$ both await an input C and output A and D . The graph obtained by plugging together $\mathcal{N}_0$ and $\mathcal{N}_1$ (matching inputs and outputs) is a DAG, whereas plugging together $\mathcal{N}_0$ and $\mathcal{N}_2$ yields a directed cycle.

Contributions & Challenges. We address this lack of compositionality and modularity by introducing methods and concepts from denotational semantics and proof theory, in a way that is fully compatible with Bayesian networks and Bayesian inference.

Our first contribution is a *compositional semantics*, equivalent to [6] but allowing to interpret and to combine modularly components. It is an adaptation of Selinger’s semantics in [10] to the factors of Bayesian networks. The technical challenge is to conciliate two different behaviours: classical variables share their values in an efficient way, whereas quantum data cannot be shared. We satisfy both requirements by introducing *quantum factors*. Strikingly, when all causes are classical they coincide with the standard factors, while in the purely quantum case they behave like tensor networks.

Our second contribution is a *typed graphical formalism*. Rather than defining yet-another-syntax, we encode quantum Bayesian networks into the graph syntax of linear logic: *proof-nets*, see Figure 2. These typed graphs allow to guarantee that composing graphs of compatible types produces a DAG, *i.e.* types ensure well-behaved compositions of systems. This formalism is *sound and complete w.r.t.* quantum Bayesian networks: every quantum Bayesian network can be represented as a proof-net, and every closed proof-net corresponds to a quantum Bayesian network. This builds upon a recent line of work connecting (classical) Bayesian networks with proof-nets [2, 4, 5]. The DAGs in Figure 4 admit the following typing: $\mathcal{N}_0 \vdash A \multimap C$, $\mathcal{N}_1 \vdash (A \multimap C) \multimap D$, $\mathcal{N}_2 \vdash C \multimap (A \otimes D)$. The DAGs $\mathcal{N}_0$ and $\mathcal{N}_1$ compose together, producing a DAG of output D , while $\mathcal{N}_2$ cannot be given any type matching the one of $\mathcal{N}_0$.

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